

Enhancing Specialty Medication Delivery Through Pharmacist-Led Integration of Subcutaneous Injection Therapy in a Neurology Clinic

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BACKGROUND

- Ocrelizumab/hyaluronidase is an FDA-approved combination of ocrelizumab, a widely used anti-CD20 B-cell depleting monoclonal antibody used as a disease-modifying therapy for treatment of multiple sclerosis (MS), and hyaluronidase, an enzyme that increases the absorption of drugs into tissue.
- Ocrelizumab/hyaluronidase is a 10-minute subcutaneous injection that can be administered in the clinic as an alternative to intravenous administration in an infusion center, allowing for increased patient convenience, decreased visit times, and reduced infusion center capacity constraints.
- The implementation of a subcutaneously administered injection process for a first-of-its-kind MS therapy in a neurology clinic has the potential to generate cost savings for both the patient and health system.

OBJECTIVES

- Our aim was to improve patient access, experience, and clinical outcomes in multiple sclerosis care by implementing an in-clinic subcutaneous treatment model as an alternative to traditional IV infusions. This approach could reduce travel and scheduling burdens, enhance operational efficiency by optimizing clinic resources, and fosters collaboration among neurology providers, pharmacists, nurses, and administrative teams. The project also aims to streamline reimbursement processes through established protocols for insurance coverage and billing, demonstrate potential cost savings for both patients and the health system, and evaluate real-world outcomes through patient-reported experience and injection administration tolerability data.

METHODS

- We identified the demand for an in-clinic subcutaneous ocrelizumab/hyaluronidase administration pathway to enhance patient convenience, reduce infusion center chair time, and improve resource utilization.
 - A multidisciplinary team—including neurology providers, nurses, specialty pharmacists, pharmacy technicians, operations staff, and patient access coordinators—was engaged with clearly defined roles for each member.
 - A standardized protocol was developed, including pre-treatment screening, patient counseling, and an emergency management plan for hypersensitivity or injection-related reactions.
 - Electronic Health Record (EHR) integration involved building the medication (ERX) in EPIC, securing formulary addition, and creating standardized order sets with dosing, pre-medication, and monitoring instructions, as well as documentation templates for nursing and pharmacist notes.
 - Staff training included targeted education on injection technique, safety monitoring, and emergency management, supported by simulation-based drills.
 - Patient access processes were established to verify insurance coverage, obtain prior authorizations, determine appropriate billing under medical or pharmacy benefits, and coordinate medication supply.
 - Clinic workflow mapping optimized patient flow from arrival through discharge, including a defined observation period and structured follow-up.
 - Patient-reported outcomes were collected at baseline, immediately post-treatment, and 72 hours post-treatment, with a planned 6-month follow-up, focusing on satisfaction, side effects, and willingness to continue therapy.

RESULTS

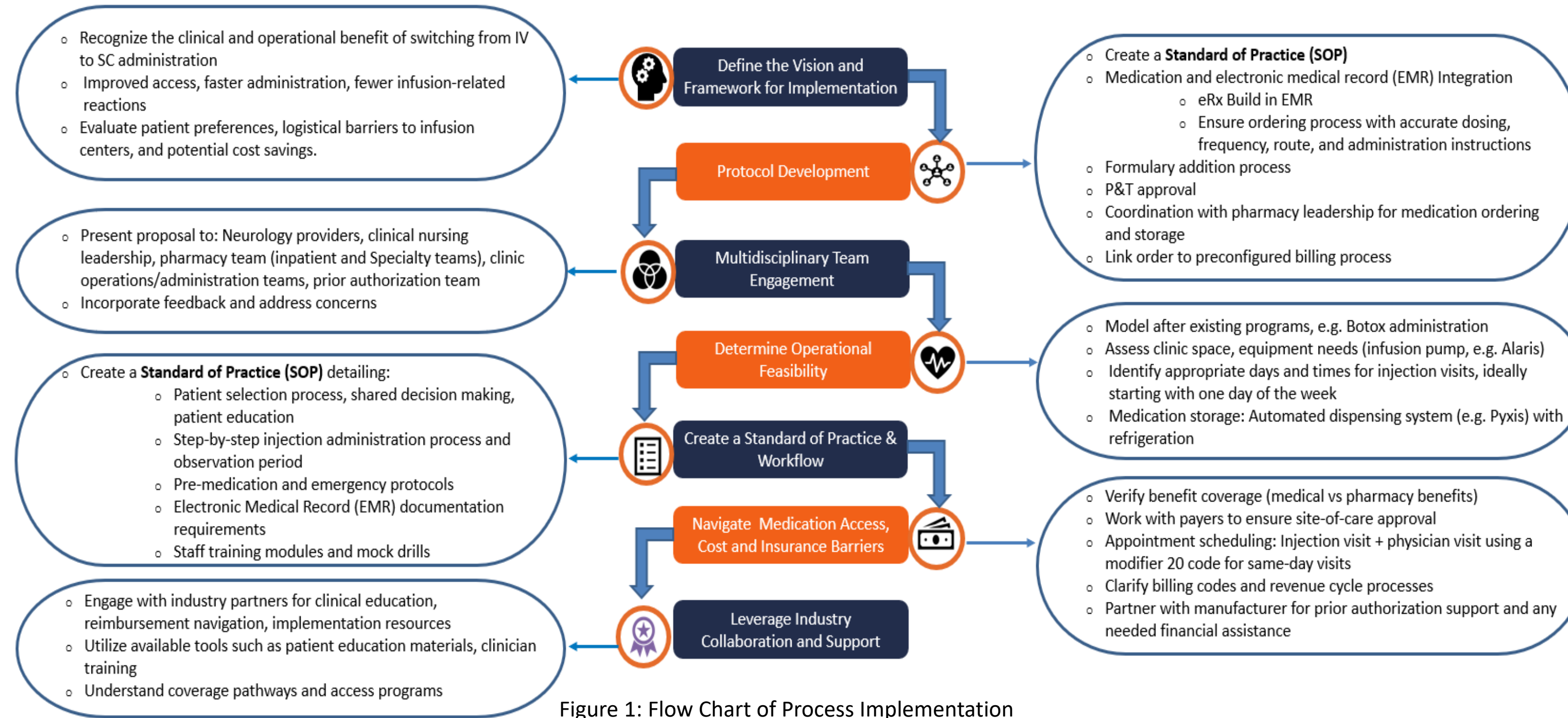


Figure 1: Flow Chart of Process Implementation

Cost Category	Intravenous Infusion (IV Center)	Subcutaneous Injection	Estimated Savings
Facility/Infusion Center Fees	High (facility fee, infusion suite charge)	Minimal (clinic visit fee)	Significant
Nursing Time	4-6+ hours RN time	30 – 60 minutes RN time	Significant
Medication Administration Time	~4 hours + pre/post monitoring	~70 – 75 min (first injection) ~25 min (subsequent injections)	Major time savings
Patient Time Commitment	~6-8 hours total (travel + infusion + monitoring)	~1-1.5 hours total (depending on travel + injection + provider visit)	Significant
Travel Costs	1 full day off work	0.5 day or less off work	Moderate
Pre-medication use	IV steroids, antihistamines, antipyretics administered at the infusion center	Oral pre-medications that can be administered before clinic visit	Mild
Monitoring Costs	Prolonged monitoring for infusion reactions	Short observation post-injection	Moderate
Risk of Infusion Reactions	Higher (especially with first dose)	Lower with subcutaneous administration	Clinical benefits
Patient Satisfaction	Lower (time burden)	Higher (convenience, comfort)	Non-monetary, high importance
Total Direct and Indirect Costs	High	Lower	Substantial

- Direct Cost Savings:** Facility fee reductions, shorter nursing time, fewer infusion center resources
- Indirect Savings:** Less time off work, less travel/childcare costs, improved patient convenience/satisfaction
- Clinical Efficiency:** Combining subcutaneous injection + provider follow-up visit into a single clinic trip prevents duplication of travel of time and costs

Table 1. Potential Impact on Cost

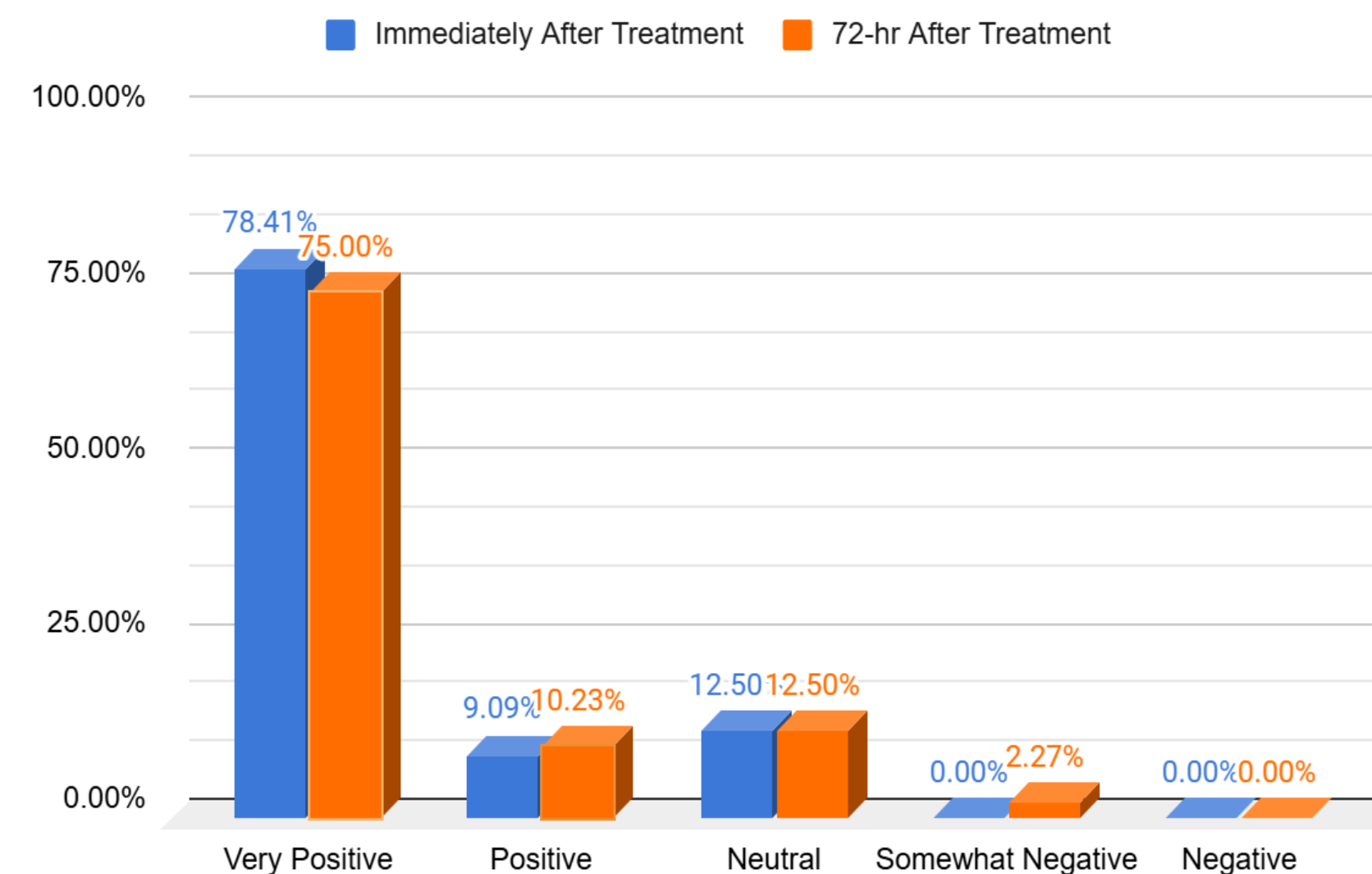
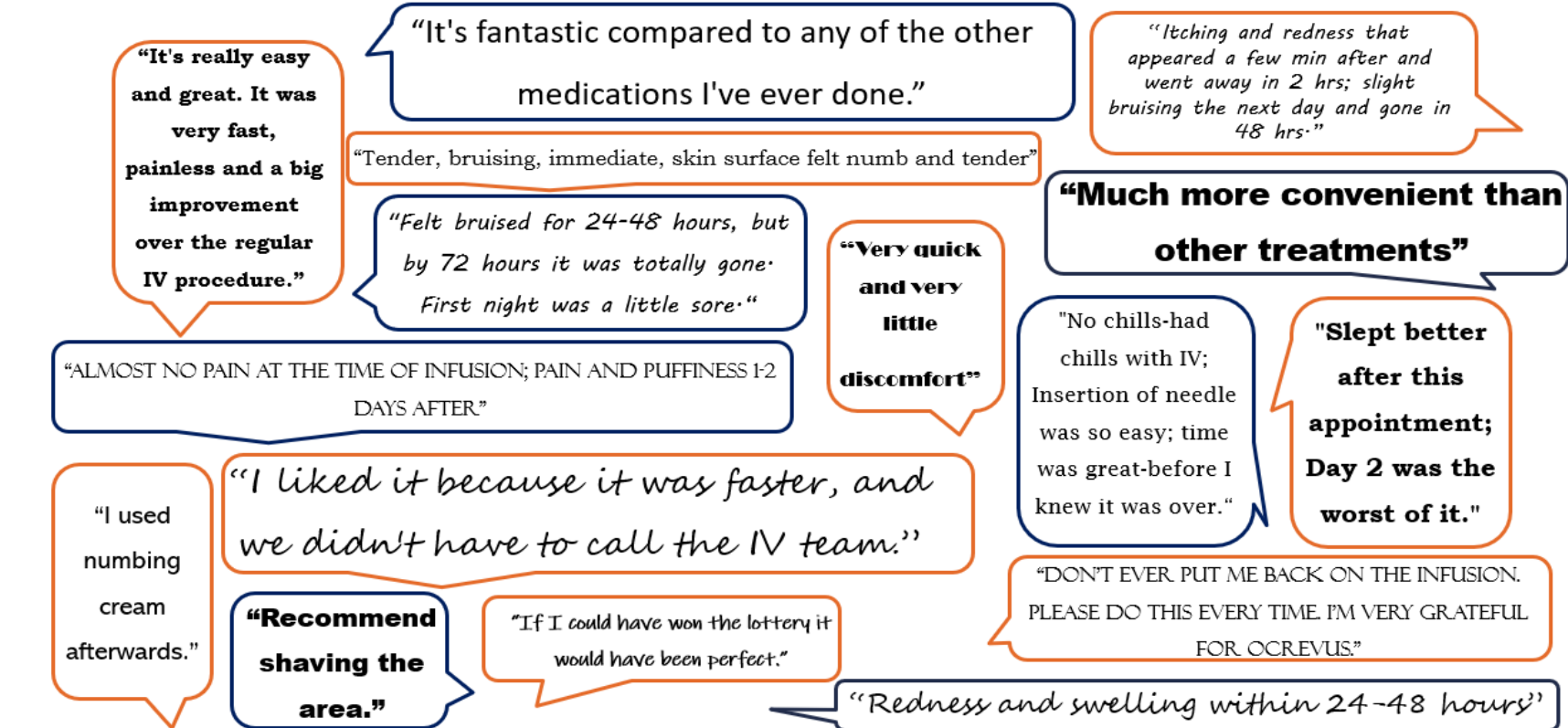


Figure 2. Patient-reported experience following treatment with ocrelizumab/hyaluronidase injection in the clinic setting

PATIENT FEEDBACK



DISCUSSION

Transitioning from infusion center-based administration to a clinic-based subcutaneous model significantly reduces treatment time, offers the potential for meaningful cost savings for patients, and enhances the overall treatment experience by delivering a highly effective therapy with minimal discomfort or side effects.

Multidisciplinary collaboration, including neurology providers, clinic nurses, pharmacy staff, administrators, and access teams, was critical to aligning workflows and refining the process for in-clinic ocrelizumab/hyaluronidase subcutaneous administration. Embedded Specialty Pharmacists played a central role by identifying gaps in access, safety, and delivery, and by leading the design and coordination of the new workflow—demonstrating the value of their clinical expertise and the potential for expanded roles within specialty pharmacy.

CONCLUSION

- A multidisciplinary approach is essential to the successful implementation of new clinic processes. Each team member brings a unique perspective, contributing valuable insights that enhance decision-making, uncover potential challenges, allowing for more collaborative, comprehensive patient-centered care.
- Clinic-embedded Specialty Pharmacists are uniquely positioned to lead the design, coordination, and implementation of innovative clinical workflows-demonstrating the integral value of their role in advancing patient care and operational efficiency within the clinic setting.
- Ocrelizumab/hyaluronidase subcutaneous injection is an effective and valuable treatment option for patients with multiple sclerosis, demonstrating high levels of patient satisfaction and preference.

REFERENCES

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- OCREVUS [ocrelizumab] Summary of Product Characteristics. Roche Pharma AG, 2022.
- Hauser SL, et al.ECTRIMS Meeting 2024;Poster P300.